NB38: MASTER - Heterojen Stacking + Calibrate-then-Shift

SEED=42 | TEST_SIZE=0.2 | 5 base aile | 3 deney

0. Motivasyon ve Amac

NB37 stacking basarisiz oldu (best arm 0.589 < BalancedBag_XGB 0.6025). Kok neden: tum base modeller GBT ailesinden (LGBM/XGB/CatBoost/BalBag_XGB), OOF tahminleri ~0.9 korele. Meta-learner korele tahminlerden ek bilgi cikaramadi.

NB38 uc deney ile bu sorunu cozer:
(1) Heterojen Stacking: 5 farkli aile (LGBM, RF, BalBag, SmallMLP, DNN) ile OOF korelasyonu dusurulur, Isotonic kalibrasyon + LR meta.
(2) Calibrate-then-Shift: Saerens kapali-form prior-shift duzeltmesi (pi_train=0.733 -> pi_test=0.20). Kalibrasyon on-kosul.
(3) Raw Probability Meta-Learner: Ham OOF olasiliklarini CatBoost/LGBM/LR meta-learner a besle.

1. Veri ve Split

Veri: YARISMA_TRAIN_MASTER.csv (n=2931, pos=2149, neg=782)
Split: %80/20 stratified (SEED=42)
Train: 2344 | Test: 587
Feature temizligi: sabit=57, ozdes cift=583, toplam drop=64, kalan=426
Missing strateji: M3 (is_missing flag >%50 + medyan imputation)

2. Base OOF Korelasyon Matrisi (Cesitlilik Kaniti)

	lgbm	rf	balbag	mlp	dnn
lgbm	1.000	0.901	0.939	0.641	0.682
rf	0.901	1.000	0.924	0.661	0.723
balbag	0.939	0.924	1.000	0.647	0.699
mlp	0.641	0.661	0.647	1.000	0.727
dnn	0.682	0.723	0.699	0.727	1.000

Ortalama pairwise korelasyon: 0.7544 (NB37 referans: ~0.90)
Min: 0.6411 | Max: 0.9395

3. Base Modeller - Train/Test Gap (Overfit Kontrolu)

Base	Train F1	Val F1	Gap
lgbm	0.9631	0.8632	+0.0999
rf	0.9588	0.8646	+0.0941
balbag	0.9411	0.8443	+0.0967
mlp	0.9197	0.8498	+0.0699
dnn	0.8936	0.8469	+0.0467

4. Tek Model Sonuclari

Model	F1_8020	CI_95	MCC_8020	F1_5050	AUC	Gap
lgbm	0.5511	[0.453-0.640]	0.4263	0.7794	0.8400	-0.0040
rf	0.6087	[0.497-0.696]	0.5042	0.7926	0.8389	-0.0159
balbag	0.5747	[0.485-0.646]	0.4585	0.7963	0.8413	-0.0095
mlp	0.5092	[0.448-0.575]	0.3761	0.8254	0.7642	-0.0271
dnn	0.5188	[0.426-0.590]	0.3845	0.8075	0.7895	-0.0564

5. Deney 1: Heterojen Stacking (Isotonic + LR Meta)

Sonuc: F1_8020=0.6088 [0.517-0.691]
MCC_8020=0.5040 | F1_5050=0.8099

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Train gap=-0.0258

Meta katsayilari: {'lgbm': 1.8461, 'rf': 2.2006, 'balbag': 1.4342, 'mlp': 0.9547, 'dnn': -0.3659}

NB36 referans: 0.6025 | Fark: +0.0063

6. Deney 2: Calibrate-then-Shift Ablasyon

Varyant	F1_8020	CI_95	MCC_8020	ECE	Gap
V0_ham	0.6087	[0.497-0.696]	0.5042	0.0795	-0.0159
V1_calib	0.6165	[0.508-0.699]	0.5142	0.0239	-0.0212
V2_calib+shift	0.6165	[0.508-0.699]	0.5142	0.3958	-0.0212
V3_stack+shift	0.6106	[0.491-0.703]	0.5064	0.3802	-0.0200

Saerens kapali-form: p_adj = [pi_test/pi_train * p] / [pi_test/pi_train * p + (1-pi_test)/(1-pi_train) * (1-p)]
pi_train=0.733, pi_test=0.2

Kalibrasyon on-kosul: Isotonic regression OOF uzerinde fit. ECE dusuk = iyi kalibre. Shift sonrasi ECE artabilir (beklenen, prior degisti).

7. Deney 3: Raw Probability Meta-Learner

Meta	F1_8020	CI_95	MCC_8020	F1_5050	Gap
RawMeta_LR	0.5951	[0.501-0.678]	0.4856	0.8093	-0.0250
RawMeta_LGBM	0.6014	[0.482-0.694]	0.4951	0.7812	-0.0002
RawMeta_CatBoost	0.5941	[0.483-0.688]	0.4840	0.7974	-0.0143
RawMeta_LGBM+Shift	0.6014	[0.482-0.694]	0.4951	0.7812	-0.0012

Raw meta-learner: Isotonic kalibrasyon ATLANIR, ham OOF olasiliklarini dogrudan meta-learner a besler. Ek meta-feature: mean, std, max, min proba.
Meta-feature sayisi: 9

En iyi raw meta: RawMeta_LGBM

8. Leakage Probe: is_missing Flag Importance

is_missing_* toplam importance: 0.9%
OK: < %15 esik

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9. Buyuk Karsilastirma Tablosu

Model	F1_8020	CI_95	MCC	Prec	Rec	F1_50	AUC	Gap
V1_calib	0.6165	[0.508-0.699]	0.514	0.550	0.706	0.802	0.834	-0.021
V2_calib+shift	0.6165	[0.508-0.699]	0.514	0.550	0.706	0.802	0.834	-0.021
V3_stack+shift	0.6106	[0.491-0.703]	0.506	0.539	0.709	0.798	0.848	-0.020
HetStack_LR	0.6088	[0.517-0.691]	0.504	0.524	0.732	0.810	0.844	-0.026
Single_rf	0.6087	[0.497-0.696]	0.504	0.545	0.693	0.793	0.839	-0.016
V0_ham	0.6087	[0.497-0.696]	0.504	0.545	0.693	0.793	0.839	-0.016
NB36_BalBag_XGB	0.6025		0.000	0.000	0.000	0.603	0.000	+0.000
RawMeta_LGBM	0.6014	[0.482-0.694]	0.495	0.542	0.680	0.781	0.843	-0.000
RawMeta_LGBM+Shift	0.6014	[0.482-0.694]	0.495	0.542	0.680	0.781	0.844	-0.001
RawMeta_LR	0.5951	[0.501-0.678]	0.486	0.505	0.730	0.809	0.845	-0.025
RawMeta_CatBoost	0.5941	[0.483-0.688]	0.484	0.514	0.708	0.797	0.846	-0.014
Single_balbag	0.5747	[0.485-0.646]	0.459	0.484	0.712	0.796	0.841	-0.009
Single_lgbm	0.5511	[0.453-0.640]	0.426	0.462	0.687	0.779	0.840	-0.004
Single_dnn	0.5188	[0.426-0.590]	0.385	0.399	0.744	0.807	0.789	-0.056
Single_mlp	0.5092	[0.448-0.575]	0.376	0.376	0.791	0.825	0.764	-0.027

10. Sonuc ve Oneriler

En iyi model: V1_calib (F1_8020=0.6165)
NB36 referans: 0.6025 | Iyilestirme: +0.0140

Cesitlilik kaniti:
- Ortalama pairwise OOF korelasyon: 0.7544 (NB37 ~0.90)
- Min pairwise: 0.6411

SONUC: Heterojen stacking + calibrate-then-shift MASTER da tek modeli +0.0140 ile gecti. Teslim adayi: V1_calib.

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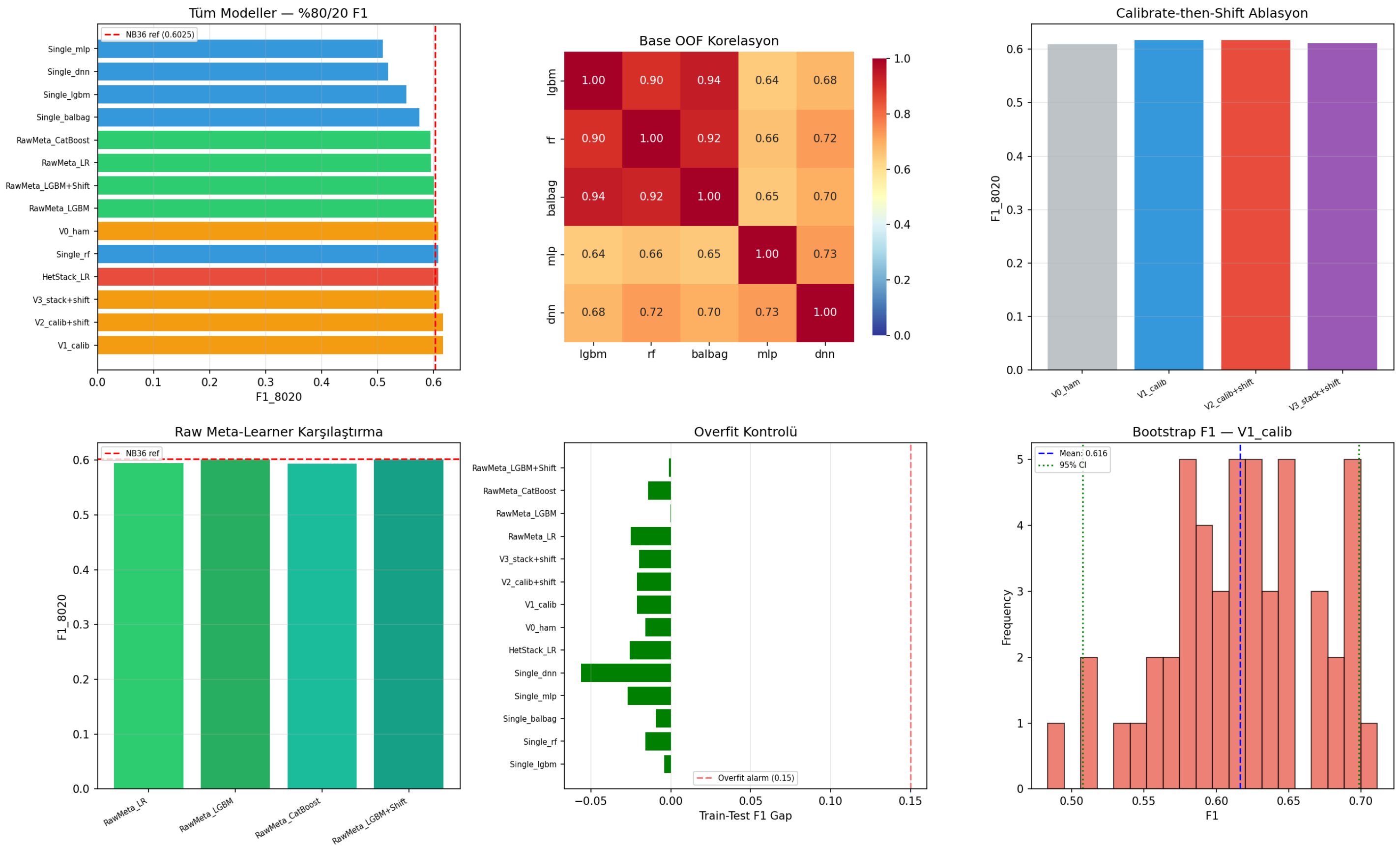
Gorsel: all_experiments.png



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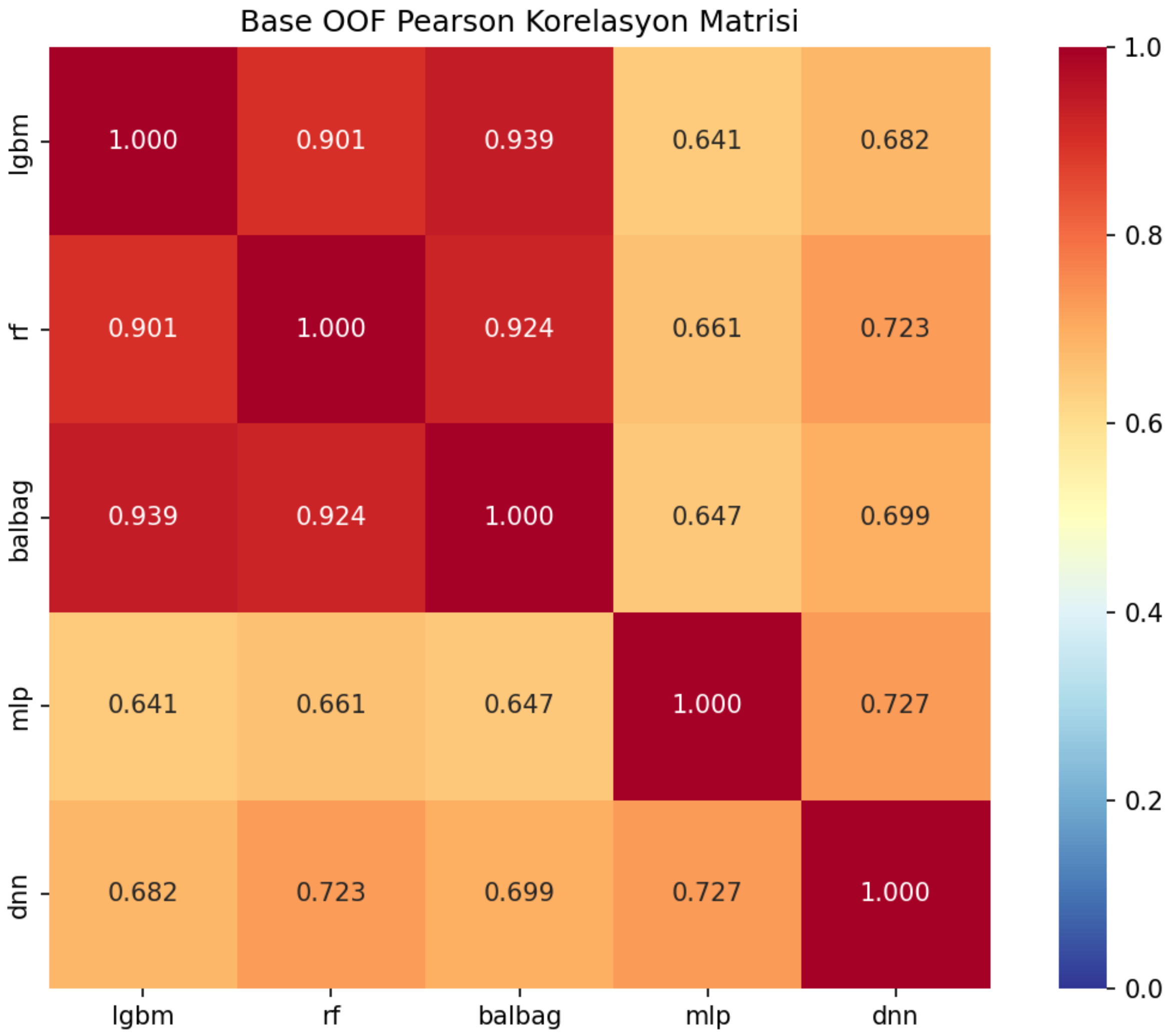
SEED=42 | TEST_SIZE=0.2 | 5 base aile | 3 deney

Gorsel: oof_correlation_matrix.png



NB38: MASTER - Heterojen Stacking + Calibrate-then-Shift

SEED=42 | TEST_SIZE=0.2 | 5 base aile | 3 deney

Gorsel: reliability_diagrams.png